

$$f(10) = 0 \cdot x_1 \oplus 0 \cdot x_2 = 0$$

$$f(11) = 0 \cdot x_1 \oplus 1 \cdot x_2 = 1$$

Step 1: $|\psi_0\rangle = |0\rangle \otimes |1\rangle$

$$|\psi_0\rangle = |00\rangle |1\rangle$$

Step 2: $|\psi_1\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$

Step 3: Apply oracle

$$|\psi_2\rangle = |\psi_3\rangle = \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

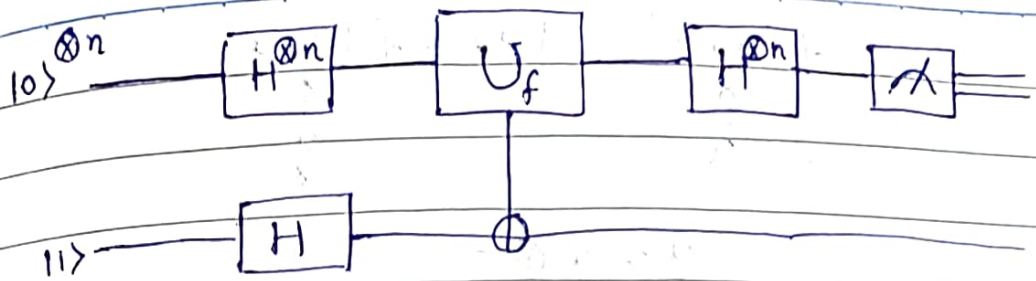
Step 4: Apply H to 1st 2 qubits.

$$|\psi_4\rangle = \frac{1}{4} (|00\rangle + |01\rangle + |10\rangle + |11\rangle - |00\rangle + |01\rangle - |10\rangle + |11\rangle + |00\rangle + |01\rangle - |10\rangle - |11\rangle - |00\rangle + |01\rangle + |10\rangle - |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$|\psi_4\rangle = |01\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$s = \underline{01}$$

Note: Bernstein - Vazirani Algorithm is special case of the Deutsch - Jozsa algorithm in which $f(x) = s \cdot x \bmod 2$.



Quantum circuit for Bernstein-Vazirani algorithm.

Simon's Algorithm :

Consider a function :

$$f : \{0,1\}^n \rightarrow \{0,1\}^n$$

such that f is 2 to 1 function.

$$f(x) = f(y) \quad ; \quad x, y \in \{0,1\}^n \text{ \& } x \neq y.$$

(each output has exactly 2 distinct inputs)

$$\text{then } y = x \oplus s, \quad s \in \{0,1\}^n \text{ \& } s \neq 00 \dots 0$$

Find the secret string s .

Example: Consider function $f : \{0,1\}^3 \rightarrow \{0,1\}^3$ defined as follows.

x	$f(x)$
000	101
001	010
010	000
011	110
100	000
101	110
110	101
111	010

with secret string $s = 110$.

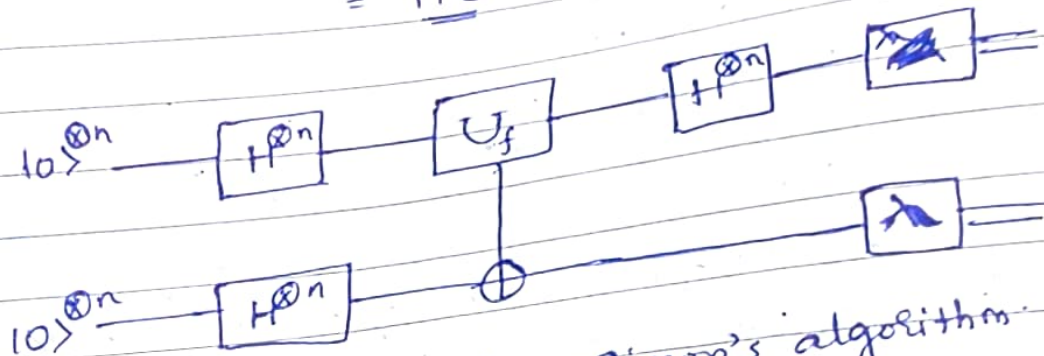
$$f(000) = 10.1 = f(110)$$

$\uparrow$ $\uparrow$
 x y

$$y = x \oplus s$$

$$= 000 \oplus 110$$

$$= \underline{110}$$



Quantum circuit for Simon's algorithm.

Quantum Fourier Transform :-

$n=1$
 $N=2$
 $|0> |1>$

The n -qubit QFT_n for $N=2^n$ is defined by

$$QFT_n : |j> \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{2\pi i j k}{N}} |k>$$

$n=2$
 $N=4$
 $|00> |01> |10> |11>$

where $j=0, 1, \dots, N-1$.

When $n=1$, we have $N=2^n=2$.

We get single qubit QFT_1 which is familiar to H-gate.

$$|j> \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^1 e^{\frac{2\pi i j k}{2}} |k> ; j=0, 1$$

case 1 : $j=0$.

$$|0\rangle \rightarrow \frac{1}{\sqrt{2}} \sum_{k=0}^1 |k\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

case 2 : $j=1$.

$$|1\rangle \rightarrow \frac{1}{\sqrt{2}} \sum_{k=0}^1 e^{i\pi k} |k\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The matrix for single qubit QFT,

$$QFT_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \left| \quad \begin{array}{l} QFT_1 \text{ is unitary,} \\ QFT_1 QFT_1^\dagger = I \end{array} \right.$$

The two-qubit QFT.

When $n=2$, we have $N=2^n=4$

We get two-qubit QFT:

$$|j\rangle \rightarrow \frac{1}{\sqrt{4}} \sum_{k=0}^3 e^{i\frac{2\pi jk}{4}} |k\rangle$$

case 1 : when $j=0$,

$$|0\rangle = \frac{1}{2} \sum_{k=0}^3 e^{i0} |k\rangle = \frac{1}{2} (|0\rangle + |1\rangle + |2\rangle + |3\rangle)$$

$$= \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\cos \frac{\pi}{2} = 0$$

$$\sin \frac{\pi}{2} = 1$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

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Case 2 : when $j = 1$

$$|1\rangle = \frac{1}{2} \sum_{k=0}^3 e^{\frac{\pi i k}{2}} |k\rangle$$

$$= \frac{1}{2} (|0\rangle + e^{\frac{\pi i}{2}} |1\rangle + e^{\pi i} |2\rangle + e^{\frac{3\pi i}{2}} |3\rangle)$$

$$e^{\frac{\pi i}{2}} = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$$

$$= 0 + i(1)$$

$$= i$$

$$e^{\pi i} = \cos \pi + i \sin \pi$$

$$= \cos \pi = -1$$

$$e^{\frac{3\pi i}{2}} = \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}$$

$$= 0 + i(-1) = -i$$

$$= \frac{1}{2} (|0\rangle + i|1\rangle - |2\rangle - i|3\rangle)$$

$$= \frac{1}{2} (|00\rangle + i|01\rangle - |10\rangle - i|11\rangle)$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ i \\ -1 \\ -i \end{bmatrix}$$

Case 3 : when $j = 2$

$$|2\rangle = \frac{1}{2} \sum_{k=0}^3 e^{\frac{\pi i k}{2}} |k\rangle$$

$$= \frac{1}{2} (|0\rangle + e^{\pi i} |1\rangle + e^{2\pi i} |2\rangle + e^{3\pi i} |3\rangle)$$

$$= \frac{1}{2} (|0\rangle - |1\rangle + |2\rangle - |3\rangle)$$

$$= \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

case 4 : when $j=3$.

$$|3\rangle = \frac{1}{2} \sum_{k=0}^3 e^{\frac{3\pi i k}{2}} |k\rangle$$

$$= \frac{1}{2} (|0\rangle + e^{\frac{3\pi i}{2}} |1\rangle + e^{3\pi i} |2\rangle + e^{\frac{9\pi i}{2}} |3\rangle)$$

$$= \frac{1}{2} (|0\rangle - i |1\rangle - |2\rangle + i |3\rangle)$$

= doubt

$$= \frac{1}{2} (|00\rangle - i |01\rangle - |10\rangle + i |11\rangle)$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ -i \\ -1 \\ i \end{bmatrix}$$

The matrix for 2-qubit QFT₂

$$QFT_2 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}$$

QFT₂ is unitary, $QFT_2 QFT_2^\dagger = I$

The inverse QFT is given by

$$QFT_n^{-1} : |k\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{\frac{-2\pi i j k}{N}} |j\rangle$$

where $k = 0, 1, \dots, N-1$.

()⁺ $\rightarrow$ Take conjugate of complex no (i) only.

1. Find the matrix for inverse QFT_2^{-1} .

We have

$$\text{QFT}_2 \text{QFT}_2^+ = I$$

$$\text{QFT}_2^{-1} = \text{QFT}_2^+$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}$$

2. Find the QFT for the function

$$|f\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Apply QFT_2 to $|f\rangle$.

$$\text{QFT}_2 |f\rangle = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 4 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \underline{|00\rangle}$$

If we apply QFT_2^{-1} to $|00\rangle$ we get?

$$QFT_2^{-1}|00\rangle = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

$$= \underline{\underline{|f\rangle}}$$

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The Search Problem:

Given unstructured database of N items of files $(y_0, y_1, \dots, y_{N-1})$

The goal is to find a specific file that satisfy a certain condition.

Given an index x , the oracle raises a flag if y_x solves the search problem.

$x: 0 \text{ to } N-1$

For given search problem we define a function

$$f: \{0,1\}^n \rightarrow \{0,1\}$$

such that $f(x) = 1$ if x solves the search problem

$f(x) = 0$ if it doesn't.

The oracle is a unitary operator defined by

$$|x\rangle |q\rangle \xrightarrow{O} -|x\rangle |q \oplus f(x)\rangle$$

where $|x\rangle$ is the index and $|q\rangle$ is an oracle qubit which is flipped if $f(x) = 1$ otherwise no change.

$$\text{If } |q\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Let us consider 2 cases:

case 1: when $f(x) = 1$,

$$|x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |x\rangle \left(\frac{|1\rangle - |0\rangle}{\sqrt{2}} \right)$$

$$= -|x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Note: $|q \oplus f(x)\rangle = \frac{|0 \oplus 1\rangle - |1 \oplus 1\rangle}{\sqrt{2}}$

$$= \frac{|1\rangle - |0\rangle}{\sqrt{2}}$$

case 2: when $f(x) = 0$,

$$|x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

($\therefore$ apply $|q \oplus f(x)\rangle$ which is $|q\rangle$)

Hence action of oracle is

$$\underbrace{|x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)}_{|q\rangle} \xrightarrow{0} (-1)^{f(x)} \underbrace{|x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)}_{|q\rangle}$$

If we neglect the oracle qubit $|q\rangle$ then action of oracle can be written as

$$|x\rangle \xrightarrow{0} (-1)^{f(x)} |x\rangle$$

Grover's Search algorithm:

Step 1: Begin $|x\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} |j\rangle$

Step 2: Apply the oracle to $|x\rangle$:

$$|x\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} (-1)^{f(x)} |j\rangle$$

Step 3: Apply QFT to $|x\rangle$

Step 4: Reverse the sign of all terms in $|x\rangle$ except for the term $|0\rangle$.

Step 5: Apply inverse QFT.

Step 6: Return to Step 2 & repeat.

Eg: Suppose we have a database contains 4 files ($N=4$) & solution is indexed by $x=2$.

Soln: Step 1: $|x\rangle = \frac{1}{\sqrt{4}} \sum_{j=0}^{4-1} |j\rangle$

$$= \frac{1}{2} (|0\rangle + |1\rangle + |2\rangle + |3\rangle)$$

Step 2: Apply oracle to $|x\rangle$

(w.k.t $|x\rangle = (-1)^{f(x)} |x\rangle$)

Since solution is indexed at $x=2$,
 $f(x)$ at 2 = 1 else 0.

$$\Rightarrow |x\rangle = \frac{1}{2} (|0\rangle + |1\rangle - |2\rangle + |3\rangle)$$

Step 3: Apply QFT_2 to $|x\rangle$

$$QFT_2 |x\rangle = \frac{1}{2} \cdot \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 2 \\ 2 \\ -2 \\ 2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \end{bmatrix} = \frac{1}{2} (|0\rangle + |1\rangle - |2\rangle + |3\rangle)$$

Step 4: Reverse sign of all terms in $|x\rangle$ except $|0\rangle$.

$$QFT_2 |x\rangle = \frac{1}{2} (|0\rangle - |1\rangle + |2\rangle - |3\rangle)$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

Step 5: Apply inverse QFT_2 i.e., QFT_2^{-1} .
(w.k.t, $QFT_2 \cdot QFT_2^{-1} = I$)

$$|x\rangle = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 0 \\ 0 \\ 4 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = |10\rangle = \underline{\underline{|2\rangle}}$$

Hence when we measure $|x\rangle$ we get solution is indexed at $x=2$.

The factoring problem:

Euclid's algorithm: The GCD is computed recursively.

$$\text{gcd}(a, b) = \begin{cases} b & \text{if } a \bmod b = 0 \\ \text{gcd}(b, a \bmod b) & \text{otherwise} \end{cases}$$

with $a > b$

Eg: $\text{gcd}(18, 12) = \text{gcd}(12, 6) = 6$.

$\text{gcd}(a, b) = 1$ if a & b co-prime.

Eg: $\text{gcd}(25, 16) = 1$.

Period finding:

Suppose a function $f(x) = f(x+r)$ then $f(x)$ is said to have period r .

x	0	1	2	3	4	5
$f(x) = 11^x \bmod 15$	1	11	1	11	1	11

This function is of period 2.

Shor's Quantum factoring Algorithm:

Step 1: Choose a random integer $a < N$, such that $\text{GCD}(a, N) = 1$.

Step 2: (Quantum Algorithm to compute period)
Create 2 quantum states of required size.

$$(N = p \cdot q)$$

and randomize them using H-gate.

Find the period r of $f(x) = a^x \bmod N$.

Step 3: If r is even and $a^{r/2} \bmod N \neq -1$ then compute $p = \gcd(N, a^{r/2} + 1)$ and $q = \gcd(N, a^{r/2} - 1)$, which are factors of N . Otherwise go back to step 1. & select different "a".

Problem:

1. Find the period r & hence find factors of $N=15$ using Shor's algorithm.

= Step 1: Choose $a = 11 < N = 15$ with $\gcd(15, 11) = 1$.

Step 2: Create 2 quantum states.

State 1: $|\Psi_1\rangle$ with size $k=3$ qubits for representing the numbers from 0 to 7 ($\leq N/2$)

State 2: $|\Psi_2\rangle$ with size $m=4$ qubits for representing the numbers from 0 to 15 ($\leq N$)

$$\text{Let } |\Psi\rangle = |\Psi_1\rangle |\Psi_2\rangle$$

Then size of $|\Psi\rangle$ is 7 (ie, 3+4).

Initialize all 7 qubits to $|0\rangle$.

$$|\Psi\rangle = |10000000\rangle = \underbrace{|1000\rangle}_{|\Psi_1\rangle} \underbrace{|10000\rangle}_{|\Psi_2\rangle}$$

Now apply H to each qubits of $|\Psi_1\rangle$

$$|\Psi_1\rangle = \left(\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right) \left(\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right) \left(\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right) |10000\rangle$$

$$= \frac{1}{\sqrt{8}} (|000\rangle + |001\rangle + |010\rangle + |011\rangle + \dots + |111\rangle) |10000\rangle$$

$|10000\rangle$

$$= \left(\frac{1}{\sqrt{8}} \sum_{k=0}^7 |k\rangle \right) |10000\rangle.$$

$$f(x) = a^x \bmod N$$

Evaluate $f(x) = 11^x \bmod 15$ for x in the state 1 simultaneously & store the result in state 2.

$$|\psi\rangle = \frac{1}{\sqrt{8}} \left(\underbrace{|10000\rangle}_{\substack{0 \\ "0"}} + \underbrace{|10001\rangle}_{\substack{1 \\ "1"}} + \underbrace{|1001\rangle}_{\substack{1 \\ "1"}} + \underbrace{|1011\rangle}_{\substack{11 \\ "11"}} + \underbrace{|1010\rangle}_{\substack{11 \\ "11"}} + \underbrace{|1001\rangle}_{\substack{11 \\ "11"}} + \dots + \underbrace{|1111\rangle}_{\substack{11 \\ "11"}} + \underbrace{|11011\rangle}_{\substack{11 \\ "11"}} \right)$$

When $x=0, f(x)=1$

$x=1, f(x)=11$

$x=2, f(x)=1$

$x=3, f(x)=11$

$x=4, f(x)=1$

$x=5, f(x)=11$

$x=6, f(x)=1$

$x=7, f(x)=11$

grouping →

$$|\psi\rangle = \frac{1}{\sqrt{8}} \left(\underbrace{|10000\rangle}_{0} + \underbrace{|1010\rangle}_{2} + \underbrace{|1100\rangle}_{4} + \underbrace{|1110\rangle}_{6} \right) \underbrace{|10001\rangle}_{1} + \left(\underbrace{|1001\rangle}_{1} + \underbrace{|1011\rangle}_{3} + \underbrace{|1101\rangle}_{5} + \underbrace{|1111\rangle}_{7} \right) \underbrace{|11011\rangle}_{11}$$

The state 1 contains the period r of interest, but only for identical values in state 2.

The period $r=2$ is the distance between the components (0, 2, 4, 6 or 1, 3, 5, 7) in the state 1 for same value in state 2 (1 or 11).

$r=2$ even & $11 \bmod 11 \neq -1$

The factors of $N=15$ are

$$p = \gcd(N, a^{r/2} + 1) = \gcd(15, 11+1) = \gcd(15, 12) = 3$$

$$q = \gcd(N, a^{r/2} - 1) = \gcd(15, 11-1) = \gcd(15, 10) = 5$$

The factors of $N=15 = p \cdot q = 3 \cdot 5$.

2. Find the period for the function $f(x) = 7^x \bmod 15$ using Shor's algorithm.

$$\gcd(N, 7) = 1$$

$$\gcd(15, 7) = 1$$

Quantum error correcting codes (QECC).

To protect quantum states against the effects of noise, we would like to develop QECC.

Difficulties in QECC:

1. No cloning. (ie. cannot copy unknown quantum state)
2. Errors are continuous.
3. Measurement destroys quantum information.

1. Three-qubit Bit Flip Code

The single bit flip error is described by

$$|0\rangle \rightarrow |1\rangle$$

$$|1\rangle \rightarrow |0\rangle$$

The Bit flip error is equivalent to action of X-gate.

Suppose Alice wants to send qubit $|\psi\rangle = a|0\rangle + b|1\rangle$ to Bob through a noisy quantum channel.

Encoding: Alice uses 3 qubits to encode one qubit. He will append 2 extra bits $|00\rangle$ to $|\psi\rangle$.

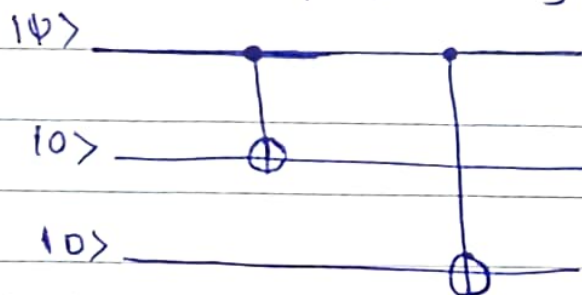
$$|\psi\rangle = |\psi\rangle |00\rangle$$

$$= (a|0\rangle + b|1\rangle) |00\rangle$$

c → control bit
T → target bit

$$= a|000\rangle + b|100\rangle \xrightarrow{\text{CNOT}} a|000\rangle + b|111\rangle$$

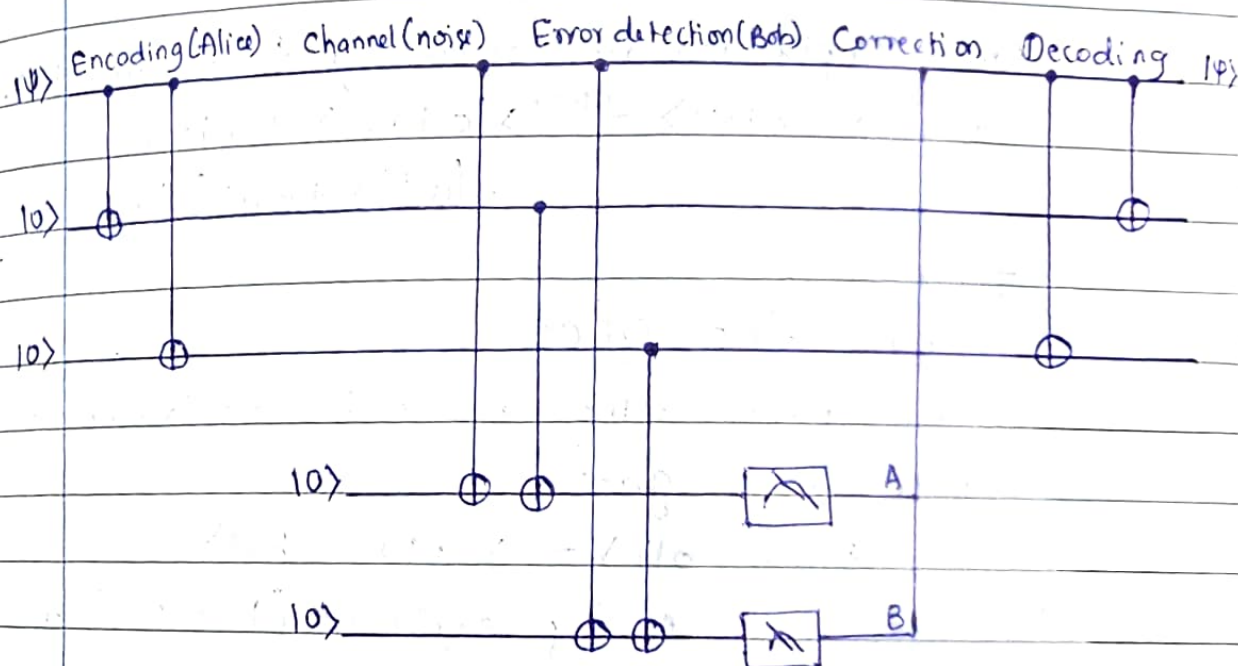
The circuit performing this encoding is



Error detection: Assuming that only a single qubit flip can occur during the transmission, Bob may receive one of the four possible states.

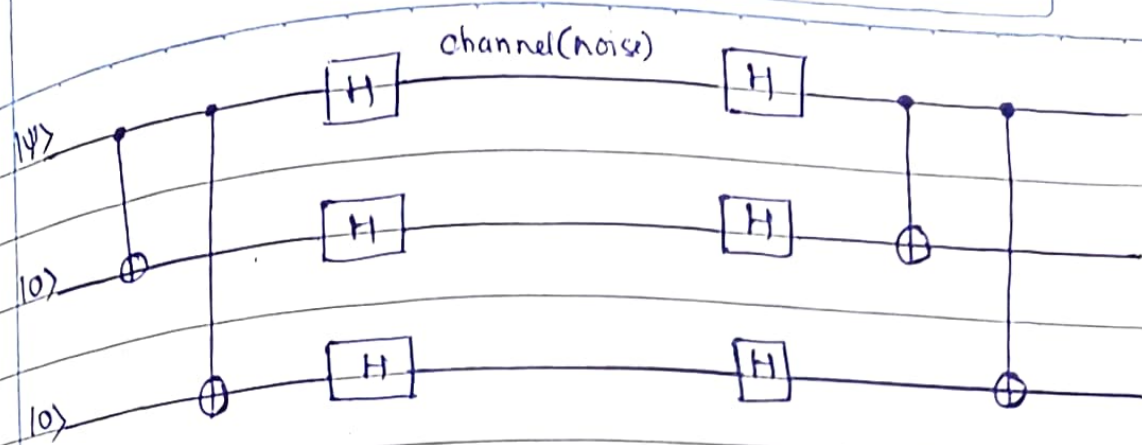
$$\begin{aligned}
 |\phi_0\rangle &= a|000\rangle + b|111\rangle & // \text{No errors} \\
 |\phi_1\rangle &= a|100\rangle + b|110\rangle & // \text{1st qubit flip} \\
 |\phi_2\rangle &= a|010\rangle + b|101\rangle & // \text{2nd qubit flip} \\
 |\phi_3\rangle &= a|100\rangle + b|011\rangle & // \text{3rd qubit flip}
 \end{aligned}$$

Now Bob has to extract from the received state which error occurred during the qubit transmission.



Now Bob has to extract from the received state which error occurred during the qubit transmission. Bob now introduces 2 extra bits $|00\rangle$ as shown in Figure 8 and applies 4 CNOT operations.

$$|-\rangle \rightarrow |+\rangle$$



1. Append extra bit : $a|0\rangle + b|1\rangle \rightarrow a|000\rangle + b|111\rangle$
2. Apply H-gate : $a|000\rangle + b|111\rangle \rightarrow a|+++ \rangle + b|--- \rangle$
3. Apply bit flip algorithm in $(|+\rangle, |-\rangle)$
4. Apply H-gate again to recover original qubit.

The Shor's nine qubit code.

It is a combination of the three qubit phase flip & bit flip codes.